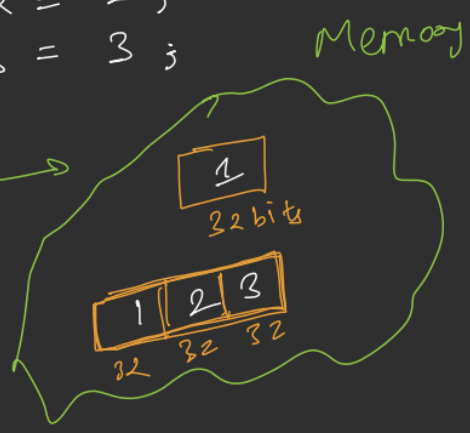
Arrays : ⇒ Collections of a data Type.

```
int rollNum1 = 1 ;
int rollNum2 = 2 ;
int rollNum3 = 3 ;
```

```
int[] rollNums ;
```

① ② ③ declaration

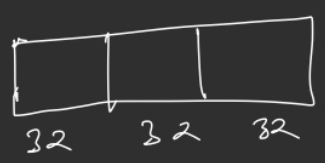



```
int[] rollNums = new int[3] ;
```

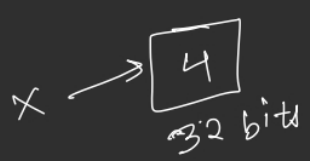
- int[] → datatype of array
- rollNums → identifier/ name of array
- new → Special Keyword ⇒ Allocate memory in heap
- [3] → Size of array

definition

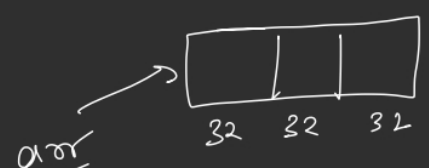
```
int[] rollNum;
rollnum = new int[3];
```




```
int x = 4;
```




```
int[] arr = new int[3];
```



r1 → 101
r2 → 102
r3 → 103

```
int[] rollNums = new int[3];
```

rollNums

101	102	103
0	1	2

```
rollNums[0] = 101;
rollNums[1] = 102;
rollNums[2] = 103;
```

```
System.out.println(rollNums[0]); // 101
                  (rollNums[1]); // 102
                  (——[2]); // 103
```

```
int[] rollNums = new int[3];
```

i i=3

101	102	103
0	1	2

x = ~~101~~ ~~102~~ ~~103~~ 104

101
102
103

```
int x = 101;
for(int i=0; i<3; i++) {
    rollNums[i] = x;
    x++;
}
```

```
for(int i=0; i<3; i++) {
    System.out.println(rollNums[i]);
}
```

101
102
103

rollNums

101	102	103
0	1	2

3

```
System.out.println(rollNums.length);

for( i=0; i< rollNums.length; i++){
        //
}
```

rollNums

101	102	103
0	1	2

103

```
System.out.println(rollNums[2]);
```

rollNums[4];

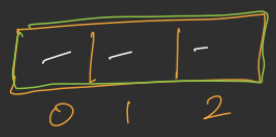
index Out of Bound Exception

Multi-Dimensional Array :

1-D

↳ Array of int

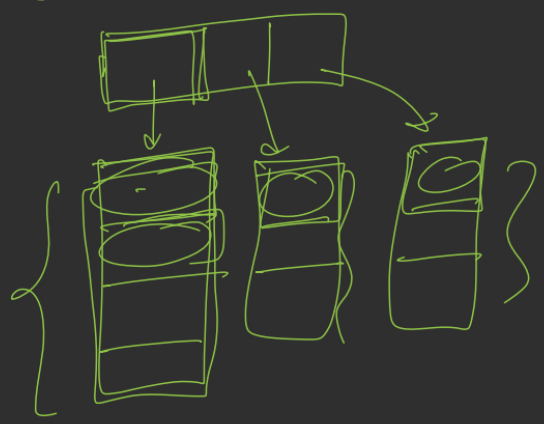
```
int [] rollNums = new int [3];
```



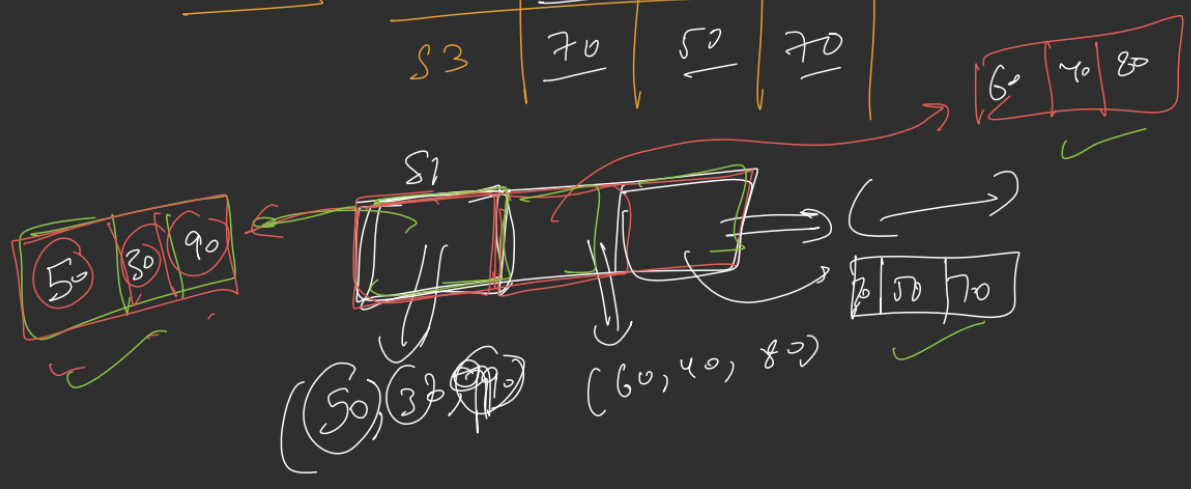
1-D Array

2-D Array :

↳ Array of Arrays



Students	Hindi	English	CSE
S1	50	30	90
S2	60	40	80
S3	70	50	70



```
int [][] marks = new int[3][3];
```

# of rows → 3, # of colums → 3

Marks →

Student ↓

	H	E	CSE
S1	50	30	90
S2	60	40	80
S3	70	50	70

(3 × 3)

	C1 (0)	C2 (1)	C3 (2)
0 r1	(0,0)	(0,1)	(0,2)
1 r2	(1,0)	(1,1)	(1,2)
2 r3	(2,0)	(2,1)	(2,2)

```
marks[0][0] = 50;
marks[0][1] = 30;
marks[0][2] = 90;
marks[1][0] = 60;
   ⋮
```

Nested loops :

Student	M	E	CSE
S1	50	30	90
S2	60	40	80
S3	70	50	70

(3×3)

```
for (int row = 0; row<3; row++){
    for (int col = 0; col<3; col++) {
        System.out.print(marks[row][col]);
    }
    System.out.println();
}
```

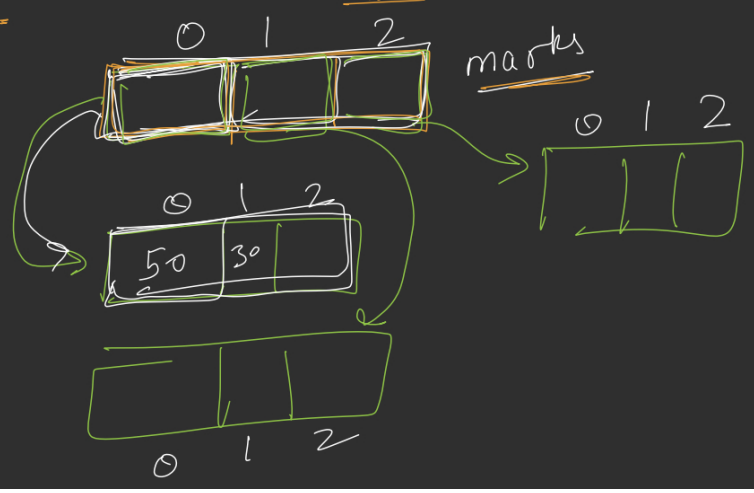
50 30 90

60 40 80

Logical Conceptual Representation of 2-D Arrays

```
int [][] marks = new int [3][3];
```

```
marks[0][0] = 50;
[0][1] = 30;
```



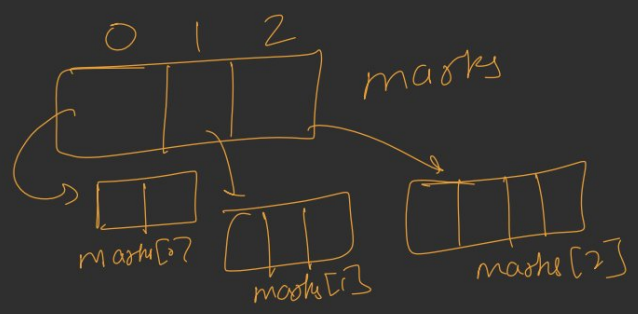
```
for (int row = 0; row < 3; row++){
    for (int col = 0; col < 3; col++) {
        System.out.print(marks[row][col]);
    }
    System.out.println();
}
```

	M	E	CSE
S1	50	30	90
S2	60	40	80
S3	70	50	70

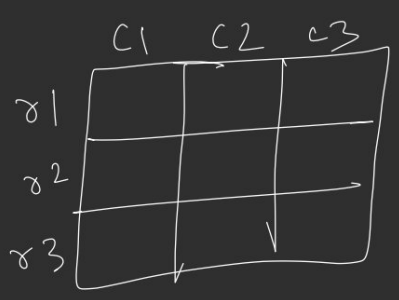
Marks → ; Student ↓ ; (3×3)

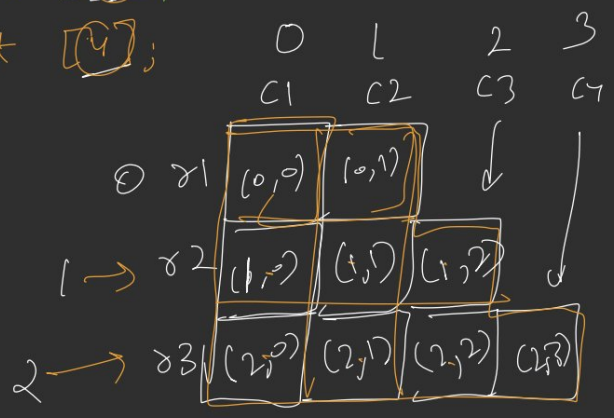
```
for (int row = 0; row < marks.length; row++){
    for (int col = 0; col < marks[row].length; col++){

    }
}
```




```
int [][] marks = new int [3][ ];
int marks[][] = new int [3][];
→ marks[0] = new int [2];
→ marks[1] = new int [3];
→ marks[2] = new int [4];
```

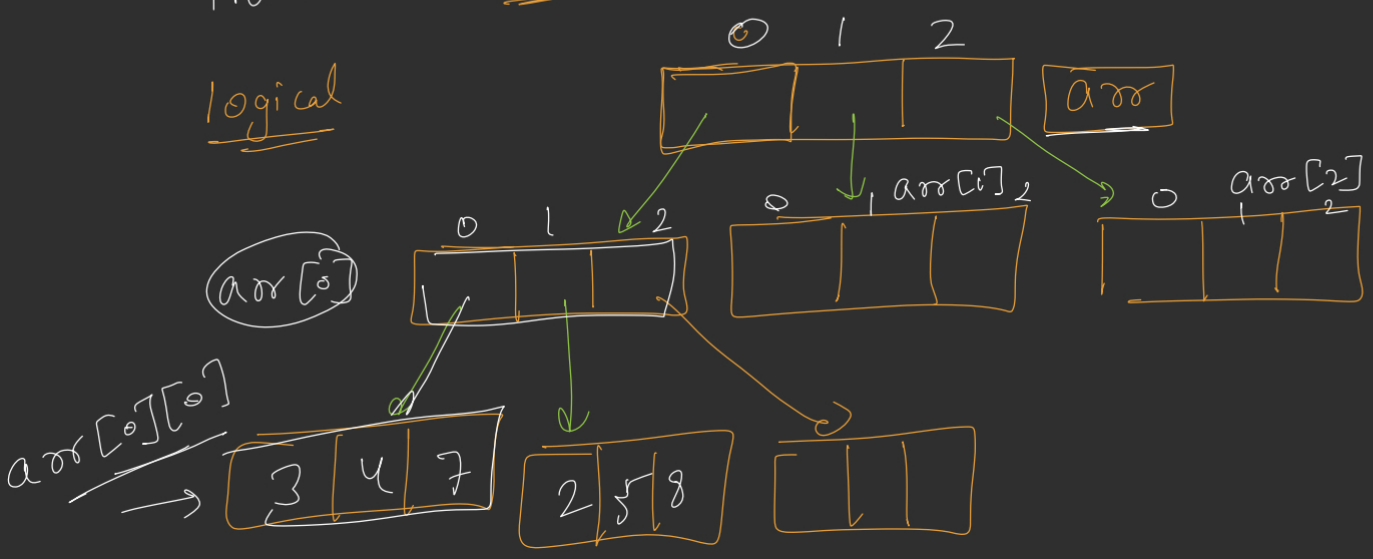


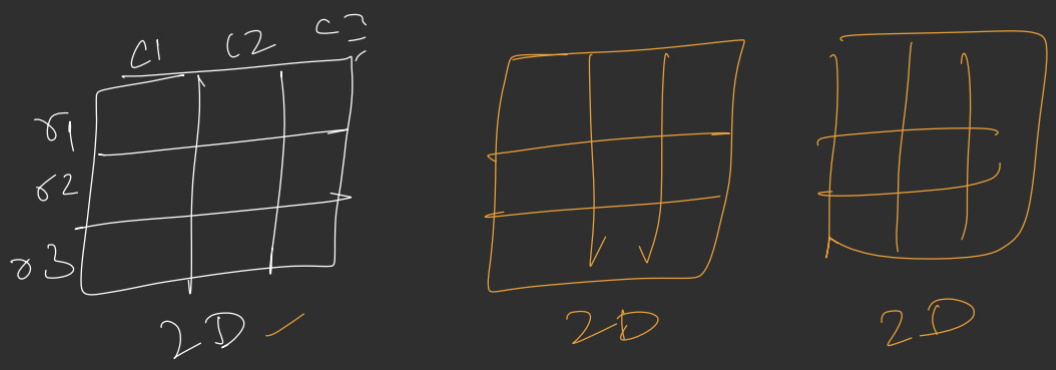


3-D Array:

```
int [][][] arr = new int [3][3][3];
```

Logical





4D, 5D, ---- ---- nD

Multi-Dimensional:

Multi-Dimensional:

1-D

```
int[] rollNums = new int[3];
```

```
rollNums[0] = 4;
rollNums[1] = 5;
```

```
int[] rollNums = {4, 5, 6};
```

```
int[][] marks = {
    {4, 5, 6}
    {7, 8, 9}
    {23, 34, 93}
};
```

Strings in Java :

```
String fname = "Aditya";
String lname = "Tandon";

System.out.println("Hello");
                   ↓
                  String
System.out.println(fname + lname);
```

"Aditya Tandon